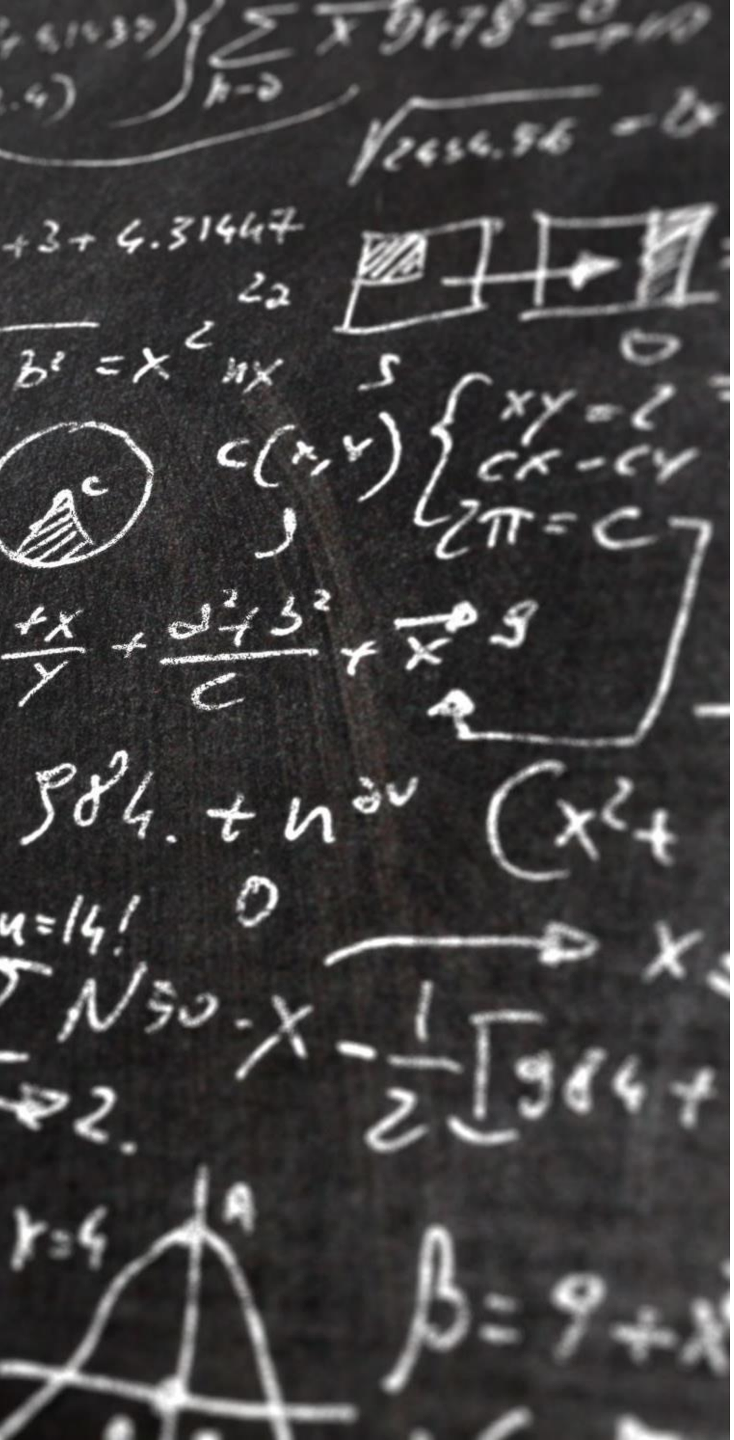


# Lecture-IV: 3-Phase System

# Energy Conversion

INSTRUCTOR: REZA SAEED KANDEZY

# Outlines

Basic concepts of STANDARD 3-phase AC circuit

- 3-phase AC circuit
  - 3-phase source
  - Circuit: current, voltage and impedance
- Basic 3-phase circuit: symmetric with balanced load

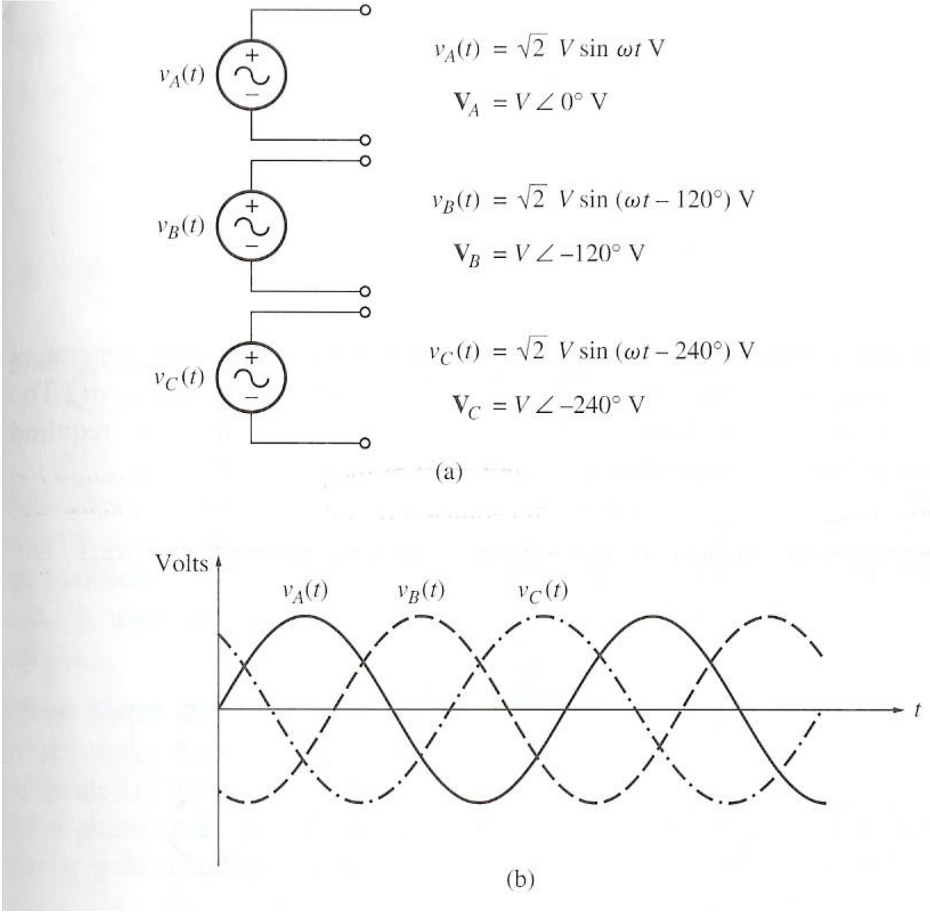
3-Phase connection

- Y connection
  - Neutral line and neutral Current
- Δ connection

Single phase representation of 3-phase circuit

# Standard Balanced Symmetric 3-Phase Circuit

# [Three Phase Voltage Source Demo]




https://www.youtube.com/watch?v=EzDs6jPrpFw

# Components of a Standard 3-Phase Circuit:

Variables:

- Voltage
- Current
- Impedance

For V or I

- Magnetite (Max value or RMS value): identical
- Frequency: identical
- Phase: strictly 120 degree apart considering the frequency adjustment)
- Phase sequence

For Impedance:

- Balanced Impedance (identical)

# BALANCED 3ϕ VOLTAGES

Balanced three phase voltages:

- same magnitude ($V_M$)
- 120° phase shift

$$v_{an}(t) = V_M \cos(\omega t)$$

$$v_{bn}(t) = V_M \cos(\omega t - 120°)$$

$$v_{cn}(t) = V_M \cos(\omega t - 240°) = V_M \cos(\omega t + 120°)$$

# BALANCED 3ϕ CURRENTS

Balanced three phase currents:

- same magnitude ($I_M$)
- 120° phase shift

$$i_a(t) = I_M \cos(\omega t - \theta)$$
$$i_b(t) = I_M \cos(\omega t - \theta - 120°)$$
$$i_c(t) = I_M \cos(\omega t - \theta - 240°)$$

# PHASE SEQUENCE

$$v_{an}(t) = V_M \cos\omega t$$
$$v_{bn}(t) = V_M \cos(\omega t - 120°)$$
$$v_{cn}(t) = V_M \cos(\omega t + 120°)$$

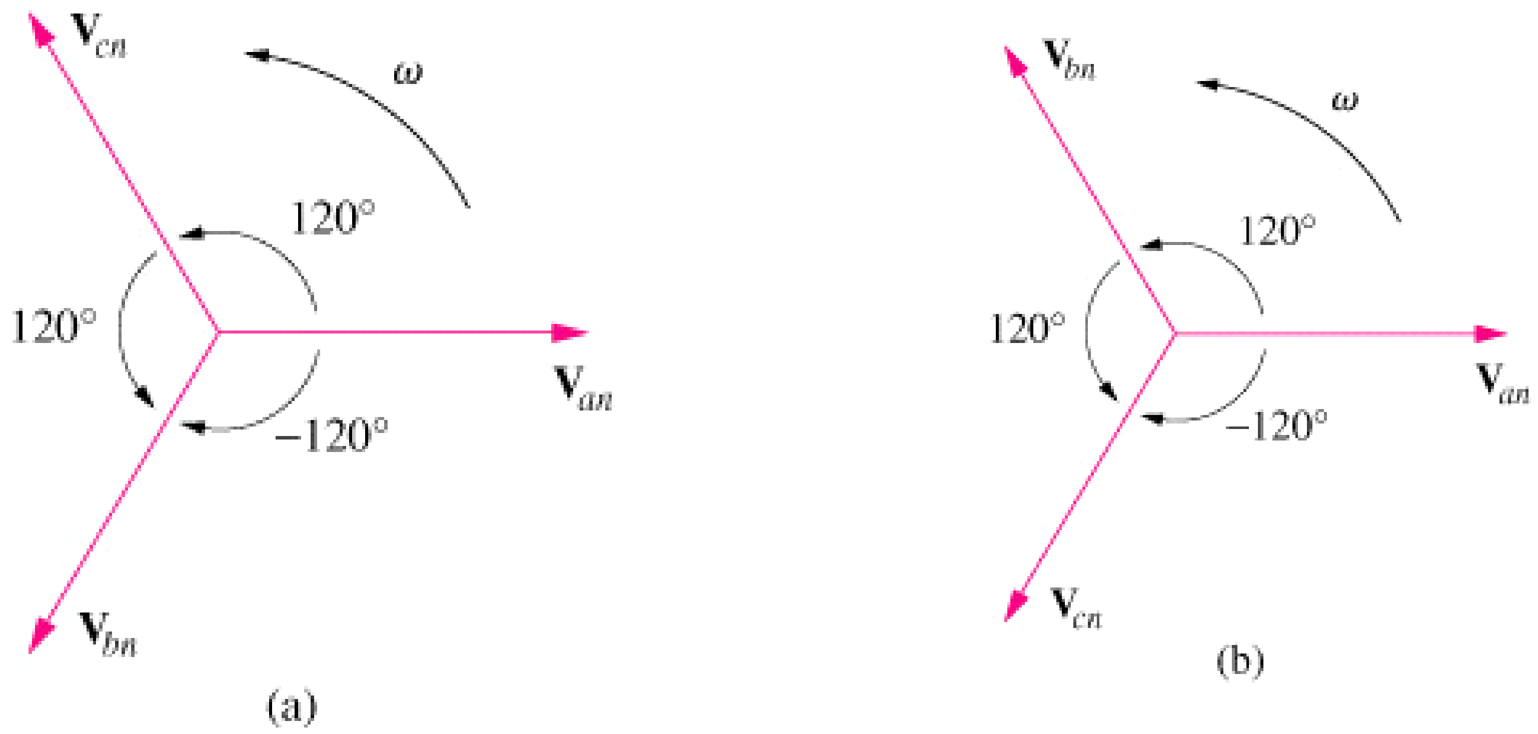
$$V_{an} = V_M \angle 0°$$
$$V_{bn} = V_M \angle -120°$$
$$V_{cn} = V_M \angle +120°$$

**POSITIVE SEQUENCE**

$$V_{an} = V_M \angle 0°$$
$$V_{bn} = V_M \angle +120°$$
$$V_{cn} = V_M \angle -120°$$

**NEGATIVE SEQUENCE**

# PHASE SEQUENCE



# EXAMPLE # 1

Determine the phase sequence of the set voltages:

$$v_{an} = 200\cos(\omega t + 10^\circ)$$
$$v_{bn} = 200\cos(\omega t - 230^\circ)$$
$$v_{cn} = 200\cos(\omega t - 110^\circ)$$

# BALANCED VOLTAGE AND LOAD

Balanced Phase Voltage: all phase voltages are equal in magnitude and are out of phase with each other by 120°.

Balanced Load: the phase impedances are equal in magnitude and in phase.

# THREE PHASE CIRCUIT

POWER

- The instantaneous power is constant

$$\boxed{\begin{aligned} p(t) &= p_a(t) + p_b(t) + p_c(t) \\ &= 3\frac{V_M\, I_M}{2}\cos(\theta) \\ &= 3V_{rms}\, I_{rms}\cos(\theta) \end{aligned}}$$

$$\boxed{\mathbf{S}_T = \mathbf{S}_A + \mathbf{S}_B + \mathbf{S}_C = 3\,\mathbf{S}_\phi}$$

# THREE PHASE QUANTITIES

| QUANTITY | SYMBOL |
| --- | --- |
| Phase current | $I_\phi$ |
| Line current | $I_L$ |
| Phase voltage | $V_\phi$ |
| Line voltage | $V_L$ |

# PHASE VOLTAGES and LINE VOLTAGES

Phase voltage is measured between the neutral and any line: line to neutral voltage

Line voltage is measured between any two of the three lines: line to line voltage.

# PHASE CURRENTS and LINE CURRENTS

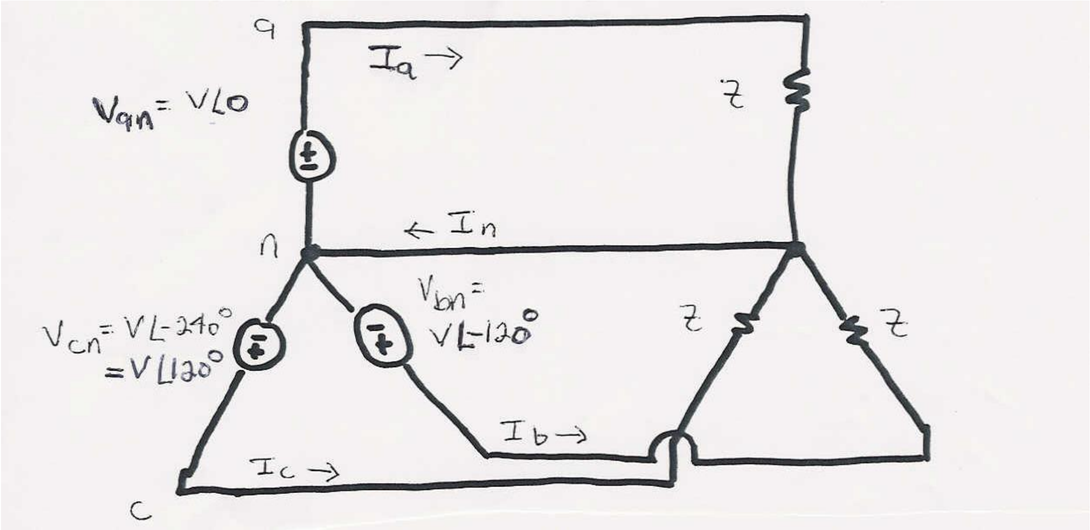
Line current ($I_L$) is the current in each line of the source or load.

Phase current ($I_\phi$) is the current in each phase of the source or load.

# Balanced 3 Phase ($\phi$) Systems

- A balanced 3 phase ($\phi$) system has
  - three voltage sources with equal magnitude, but with an angle shift of $120^\circ$
  - equal loads on each phase
  - equal impedance on the lines connecting the generators to the loads
- Bulk power systems are almost exclusively $3\phi$
- Single phase is used primarily only in low voltage, low power settings, such as residential and some commercial

# Balanced 3ϕ -- No Neutral Current




$$I_n = I_a + I_b + I_c$$

$$I_n = \frac{V}{Z}(1\angle 0° + 1\angle -120° + 1\angle 120°) = 0$$

$$S = V_{an}I_{an}^* + V_{bn}I_{bn}^* + V_{cn}I_{cn}^* = 3V_{an}I_{an}^*$$

# Symmetric-balanced Three-phase Circuit

Symmetric three phase voltage source

Balanced three phase impendence

- Each of the voltage source could be connected to one of three identical loads by a pair of wire.
- The currents which flows in the three phases ($I_A$, $I_B$, $I_C$) can be found from the equation

$$I = \frac{V}{Z}$$

Benefit:

- Simplify the computation
- Taking the advantage of the characteristics of the current in Neural line (A common line).

→ Only <u>*four*</u> wires are needed to supplying 3Φ-power.

# Advantages of 3ϕ Power

- Can transmit more power for same amount of wire (twice as much as single phase)
- Torque produced by 3ϕ machines is constrant
- Three phase machines use less material for same power rating
- Three phase machines start more easily than single phase machines

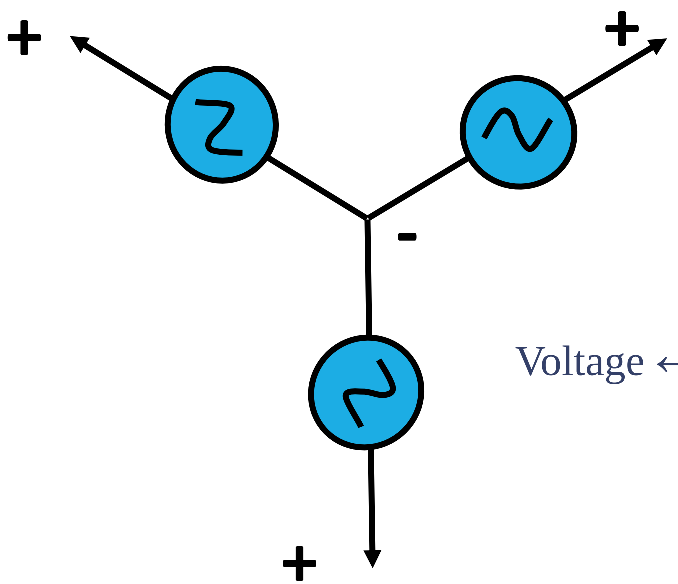
# Three-phase Connection

Structure / Connection

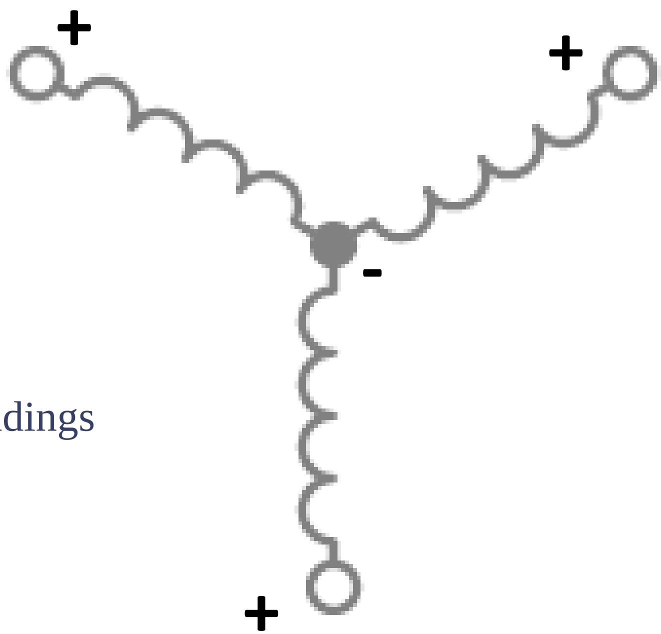
Quantitative analysis

# The Wye Connection

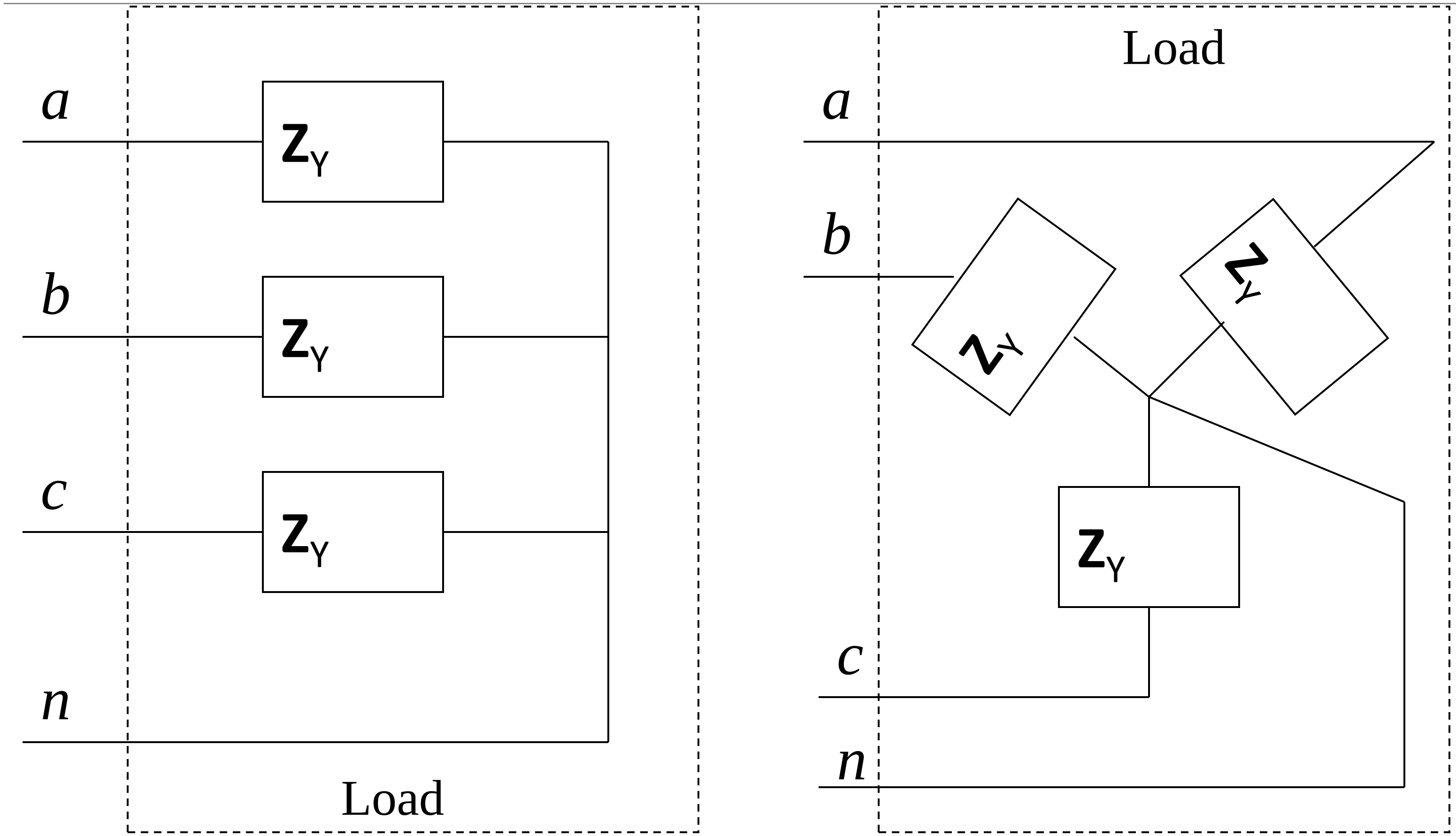
- The wye connection is made by connecting one end of each of the three-phase windings together as shown




Voltage ↔ Windings

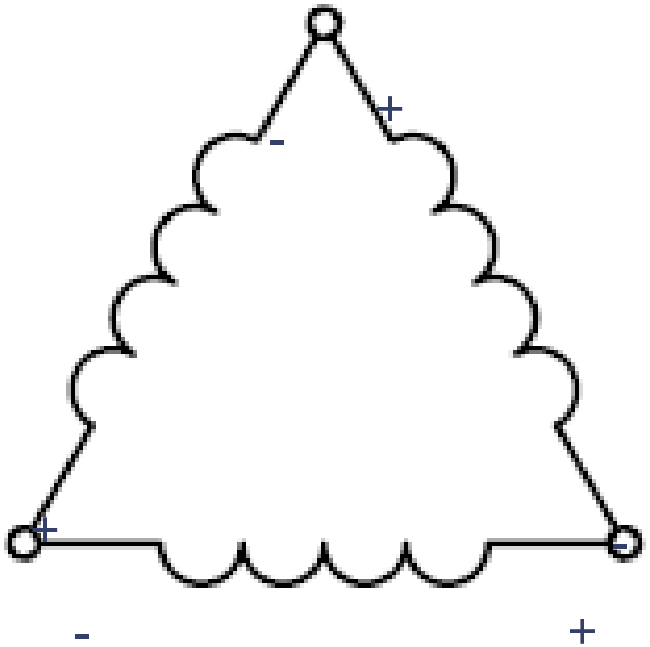


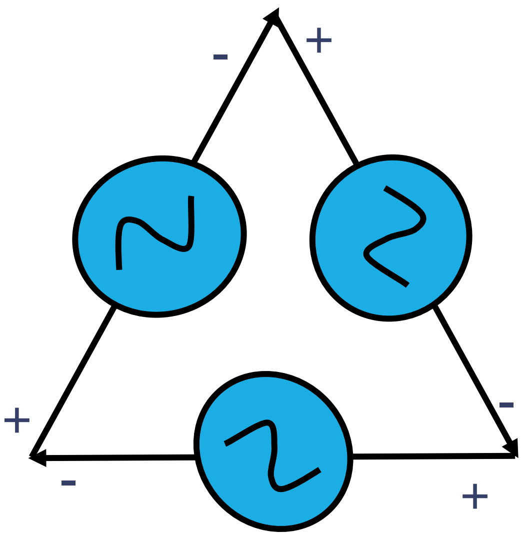
# Wye (Y) Connected Load



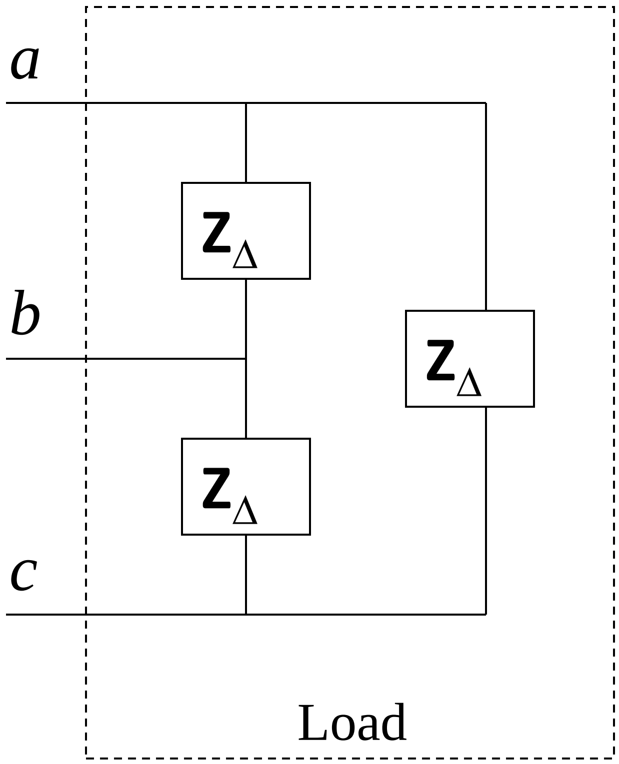
# The Delta Connection

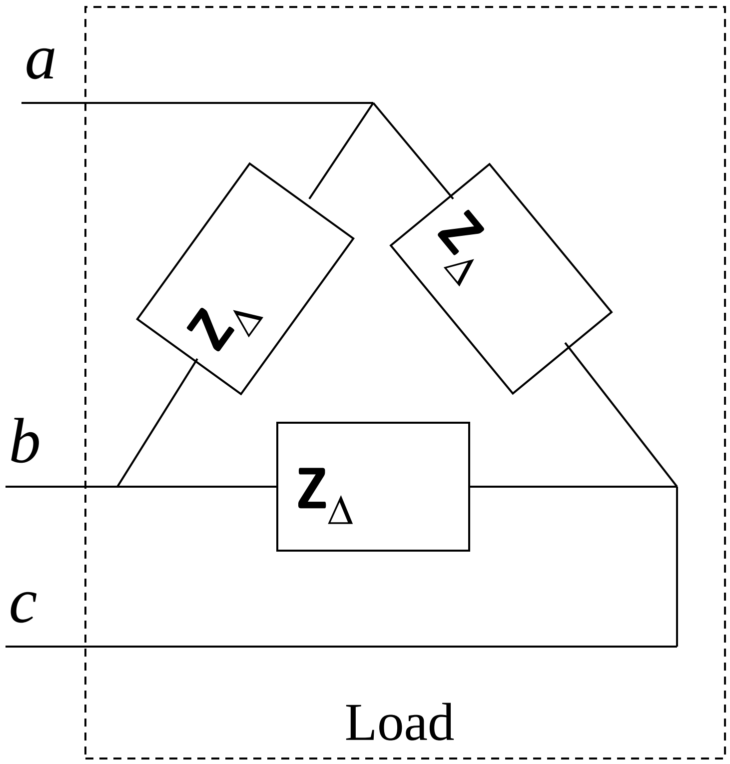
- The delta connection is made by connecting each of the three-phase windings in a triangular shape, like the Greek letter delta (Δ)



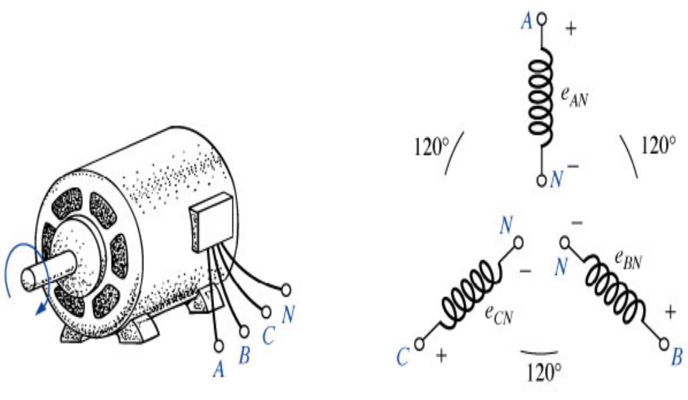



# Delta (Δ) Connected Load

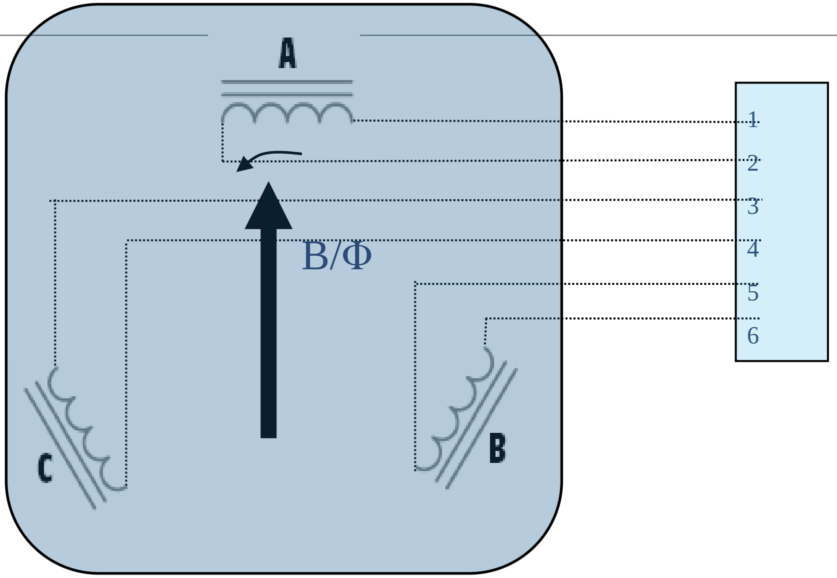


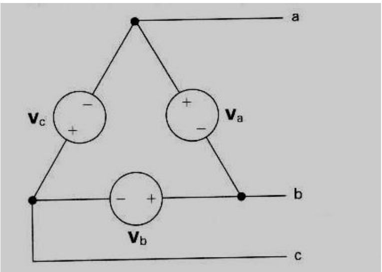



# Connections

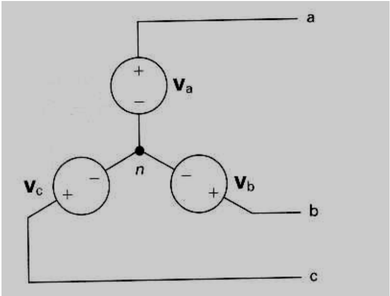


(a) (b)



(b)



(a)

# Three phase System:
# Connections of Source and Load

Both the three phase source and the three phase load can be connected either Wye or DELTA.

We have 4 possible connection types. Y-Y Y-Δ Δ-Δ Δ-Y

Balanced Δ connected load is more common.

Y connected sources are more in common.

# Per Phase Analysis

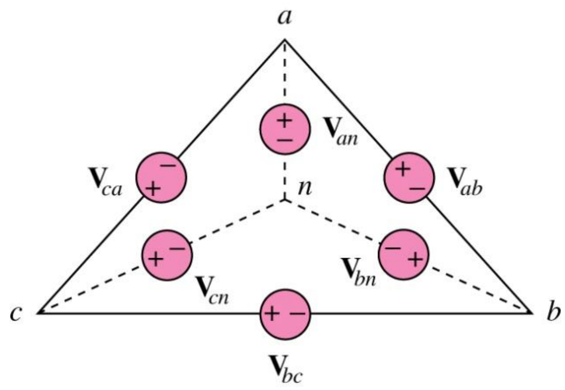
- Per phase analysis allows analysis of balanced $3\phi$ systems with the same effort as for a single phase system
- Balanced $3\phi$ Theorem: For a balanced $3\phi$ system with
  - All loads and sources Y connected
  - No mutual Inductance between phases
- Then
  - All neutrals are at the same potential
  - All phases are COMPLETELY decoupled
  - All system values are the same sequence as sources. The sequence order we've been using (phase b lags phase a and phase c lags phase a) is known as "positive" sequence.

# Per Phase Analysis Procedure

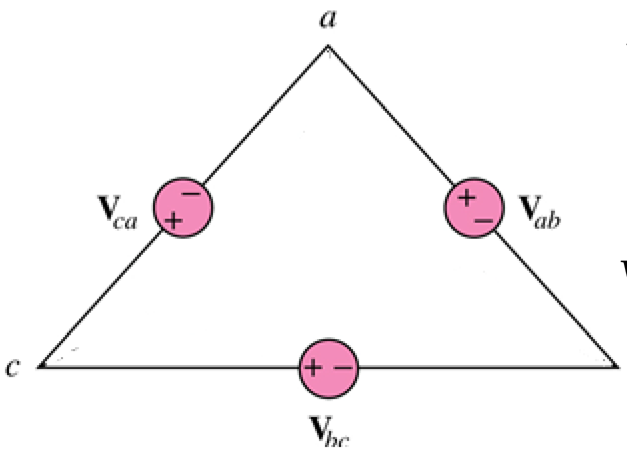
1. Convert all Δ load/sources to equivalent Y's
2. Solve phase "a" independent of the other phases
3. Total system power $S = 3\ V_a\ I_a^{*}$ (when to use sqrt(3)?)
4. If asked, phase "b" and "c" values can be easily determined (i.e., ±120° degree phase shifts depending on the phase sequence)
5. If necessary, convert back to the original circuit by determining line-line values or internal Δ values based on Line-Phase relations.

# Converting Delta-Connected Source to Wye-Connected Source !

**Transforming a Delta connected source to an equivalent Wye connection**

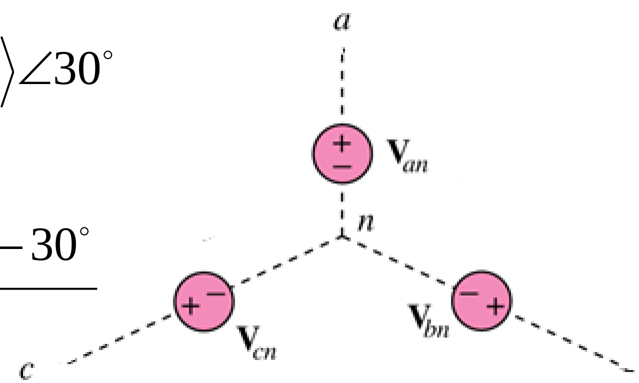



**Transforming a Wye connected source to an equivalent Delta connection**



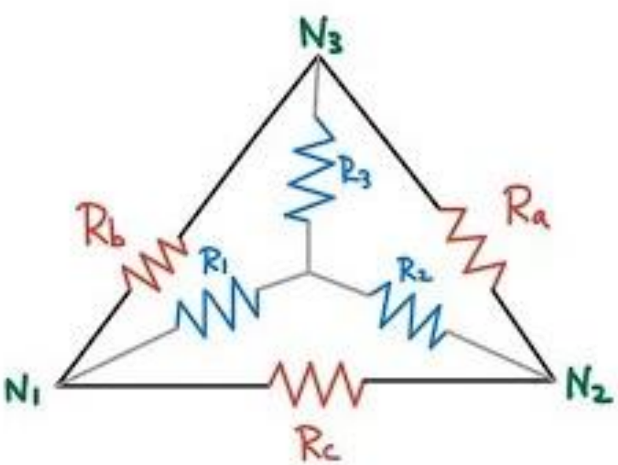

$$V_P^{delta} = \sqrt{3}\left\langle V_{an}^{Y-EQ} \right\rangle \angle 30^\circ$$

$$V_{an}^{Y-EQ} = \frac{\left\langle V_P^{delta} \right\rangle \angle -30^\circ}{\sqrt{3}}$$



# Wye-Delta Impedance Conversion

General rule:




$Y \leftarrow \Delta$

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}$$

$$R_2 = \frac{R_a R_c}{R_a + R_b + R_c}$$

$$R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

$\Delta \leftarrow Y$

$$R_a = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$R_b = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2}$$

$$R_c = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3}$$

# Wye-Delta Impedance Conversion

A **balanced load** is one in which the phase impedances are equal in magnitude and in phase.

For a balanced wye connected load:

$$Z_1 = Z_2 = Z_3 = Z_Y$$

For a balanced delta connected load:

$$Z_a = Z_b = Z_c = Z_\Delta$$

$$\boxed{Z_\Delta = 3Z_Y} \quad \text{or} \quad \boxed{Z_Y = \frac{1}{3} Z_\Delta}$$

# Examples

# Example: phase sequence

Determine the phase sequence of the set of voltages

$v_{an} = 200 \cos(\omega t + 10^\circ)$

$v_{bn} = 200 \cos(\omega t - 230^\circ)$, $v_{cn} = 200 \cos(\omega t - 110^\circ)$

**Solution:**

The voltages can be expressed in phasor form as

$$V_{an} = 200\angle 10^\circ V \qquad V_{bn} = 200\angle -230^\circ V \qquad V_{cn} = 200\angle -110^\circ V$$

We notice that $\mathbf{V}_{an}$ leads $\mathbf{V}_{cn}$ by $120^\circ$ and $\mathbf{V}_{cn}$ in turn leads $\mathbf{V}_{bn}$ by $120^\circ$.

Hence, we have an *acb* sequence.

# Example: Phase sequence

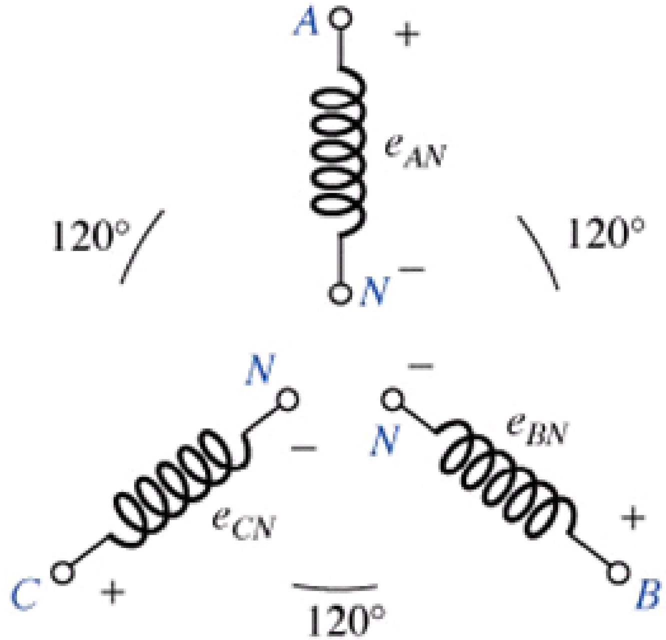
Given that $V_{bn} = 110\angle 30° V$ find **V***an* and **V***cn*, assuming a positive (*abc*) sequence.

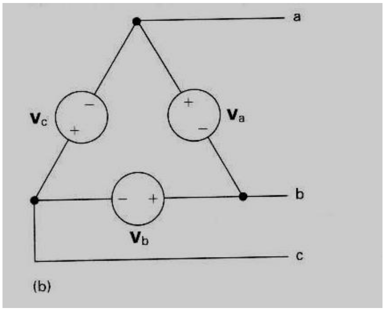
$V_{an} = 110\angle 150° V$

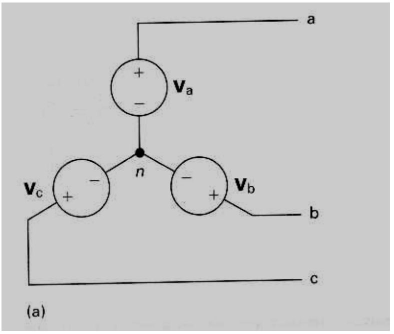
**Answer:**

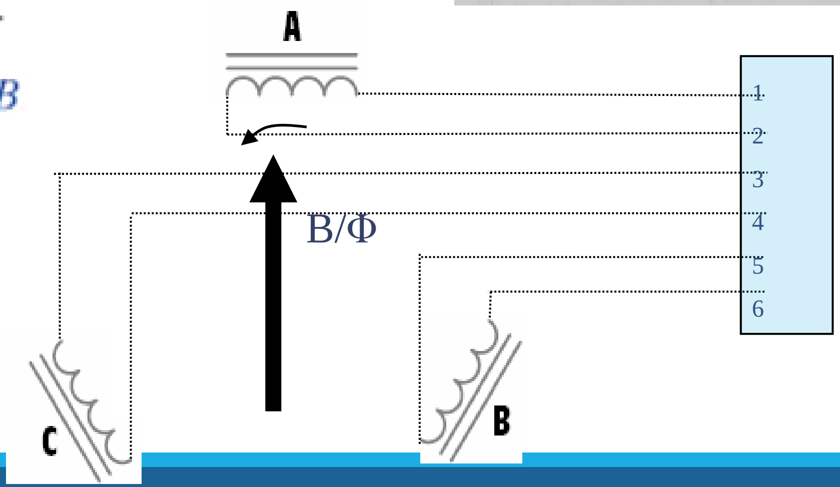
$$V_{cn} = 110\angle -90° V$$

# Example: 3-phase connection

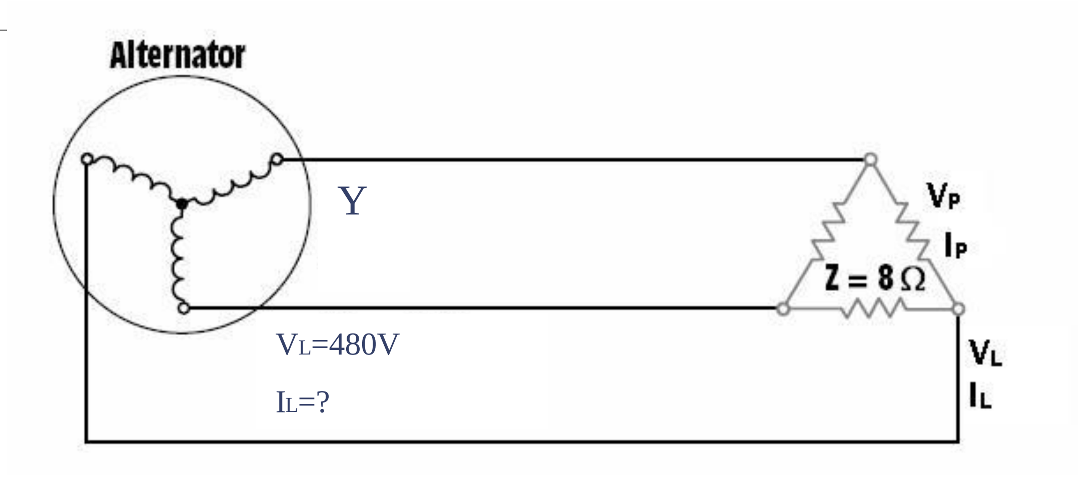


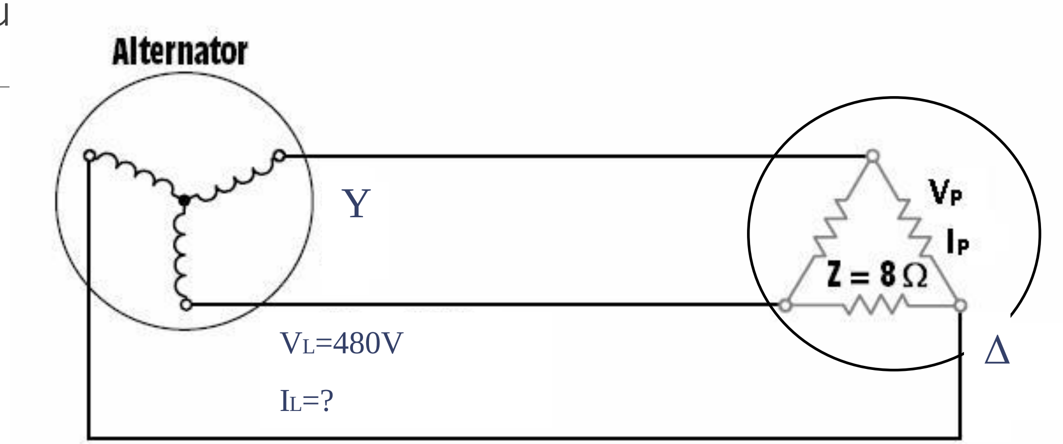
# Example: Line Quantity and Phase Quantity




A wye-connected three-phase alternator supplies power to a delta-connected resistive load. The alternator has a line voltage of 480 V. Each resistor of the delta load has 8 Ω of resistance.

# Simple 3-phase analysis

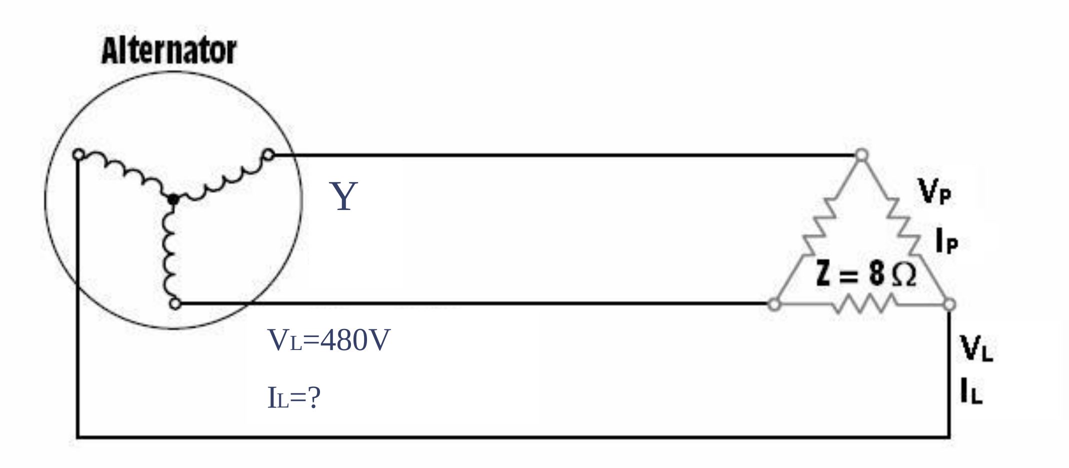
(Using Line Qu




Find the following values (rms):

- $V_{L(Load)}$ – line voltage of the load
- $V_{P(Load)}$ – phase voltage of the load
- $I_{P(Load)}$ – phase current of the load
- $I_{L(Load)}$ – line current delivered by the alternator
- $I_{L(Alternator)}$ – phase current of the alternator
- $V_{P(Alternator)}$ – phase voltage of the alternator
- $P$ – true power

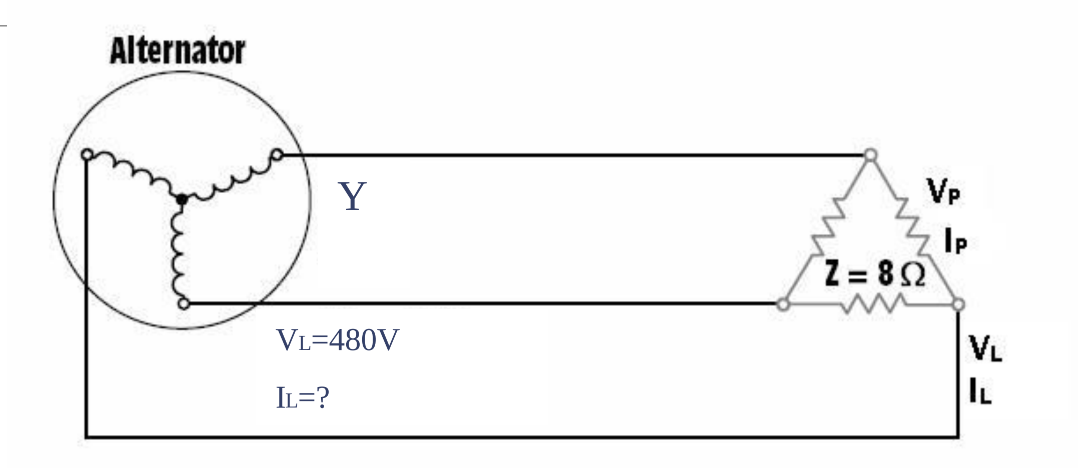
# Example: Simple 3-phase analysis, con’t




The resistors are connected directly across the three-phase source. This means that the line voltage from the source is the same as the line voltage across the load.

$$V_{L(Load)} = 480\text{ V}$$

# Example: Simple 3-phase analysis, con’t

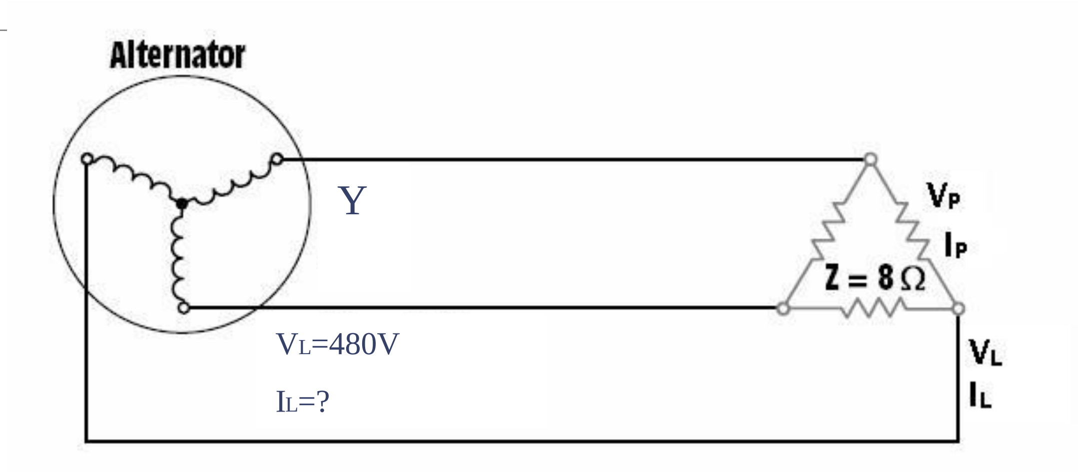



The three resistors of the load are connected in a delta connection. In a delta connection, the phase voltage is the same as the line voltage.

$$V_{P(Load)} = V_{L(Load)}$$

$$V_{P(Load)} = 480 \text{ V}$$

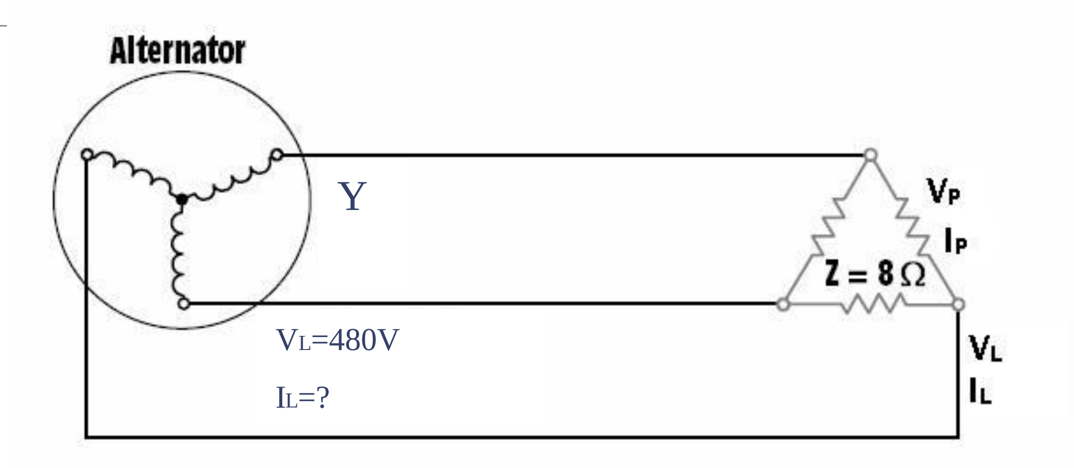
# Example: Simple 3-phase analysis, con't




We know that the resistors are each one phase of the load. Since we know the phase voltage, we can calculate the phase current using Ohm's Law:

$$I_{P(Load)} = \frac{V_{P(Load)}}{Z} = \frac{480}{8} = 60A$$

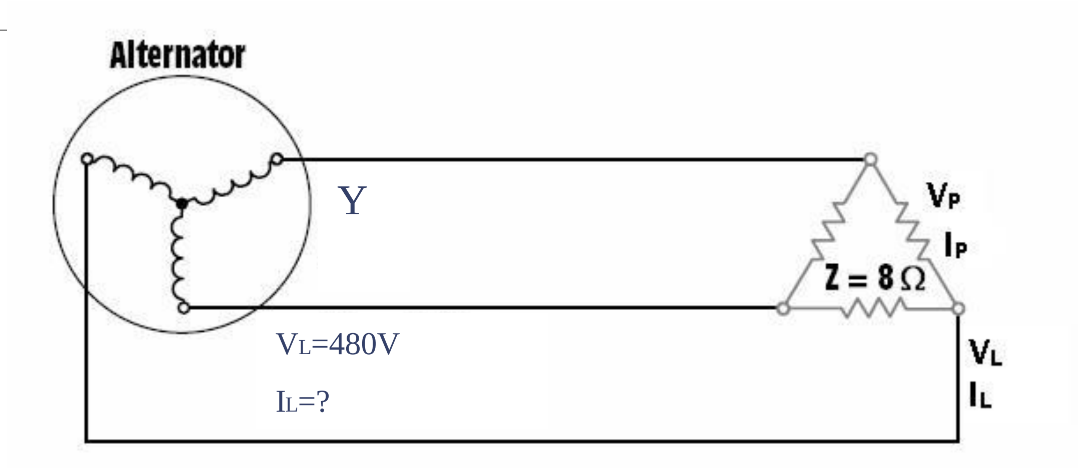
# Example: Simple 3-phase analysis, con't




The line current supplying the delta connection must be 1.732 times larger than the phase current through each resistor, so now we know enough to calculate the line current flow in each load.

$$I_{L(Load)} = I_{P(Load)} \text{ X } 1.732 = 60 \text{ X } 1.732 = 103.92 \text{ A}$$

# Example: Simple 3-phase analysis, con't

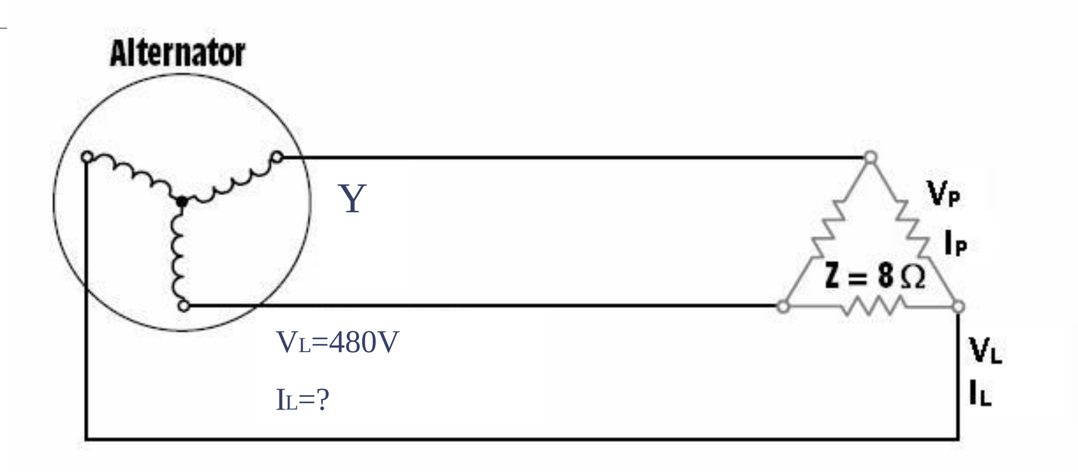



The three-phase source must supply this current, so the line current from the alternator must be the same as the line current of its load.

The source is set up in a wye connection, so we know that the phase current and line current must be the same.

$$I_{L(Alt)} = 103.92 \text{ A}$$

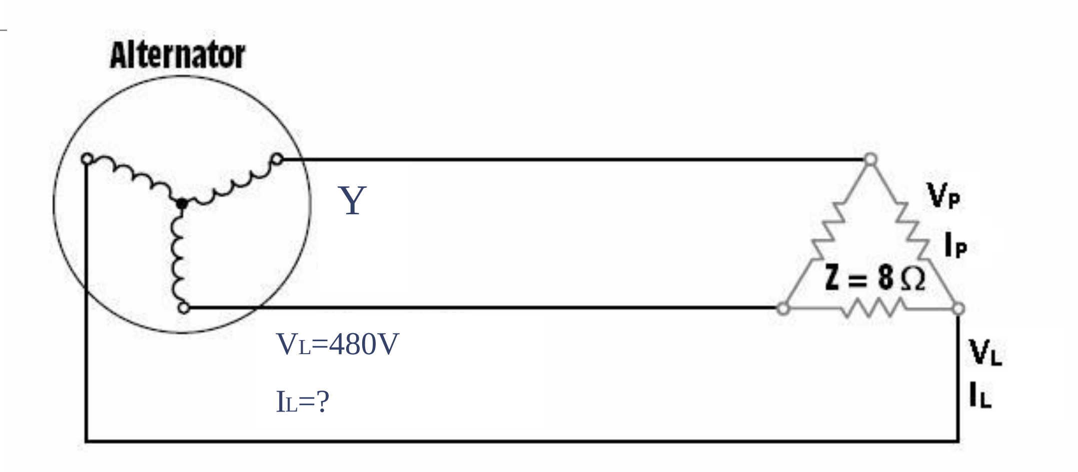
# Example: Simple 3-phase analysis, con't




The phase voltage of a wye connection is smaller than the line voltage. We must divide this known value by the 1.732 in order to get the phase voltage of the alternator:

$$V_{P(Alt)} = \frac{V_{L(Alt)}}{1.732} = \frac{480}{1.732} = 277.13V$$

# Example: Simple 3-phase analysis, con’t




In this circuit, the load is purely resistive, so the all of the power dissipated is real power.

$$P = 1.732 \times V_{L(\text{Alternator})} \times I_{L(\text{Alternator})}$$

$$P = 1.732 \times 480 \times 103.92$$

$$P = 86{,}394.93\ \text{W}$$

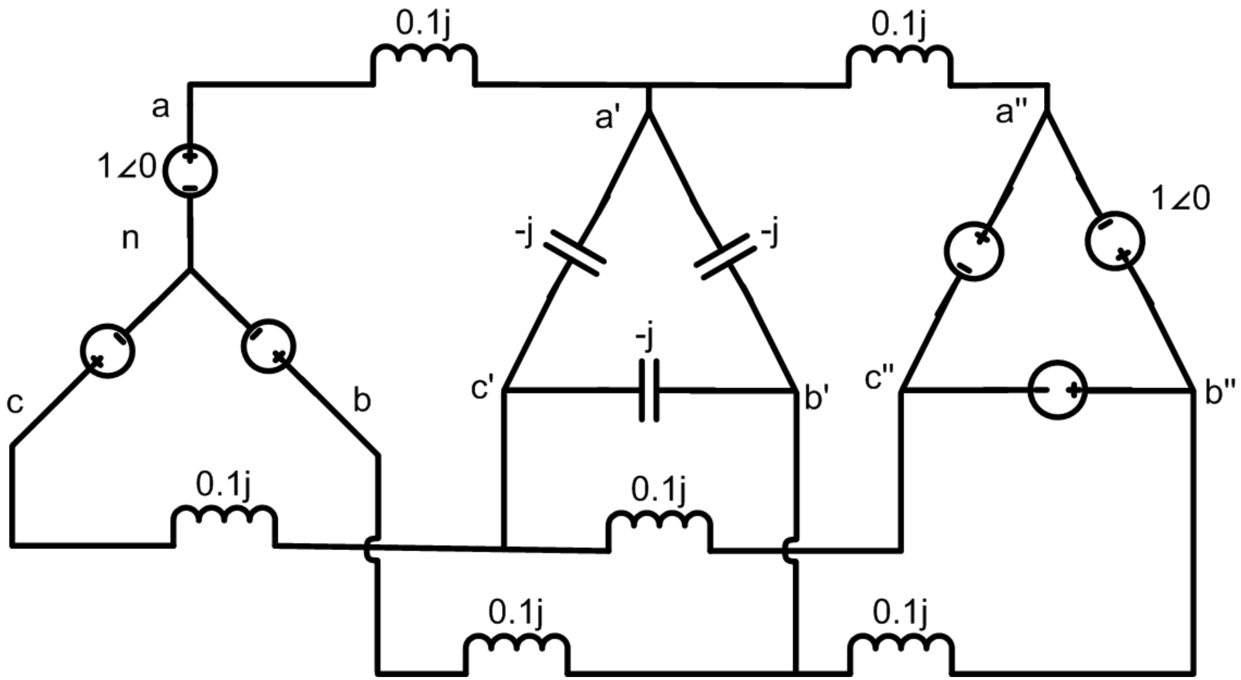
# Example: Standard Per Phase Analysis

Assume a $3\phi$, Y-connected generator with $V_{an} = 1\angle 0°$ volts supplies a $\Delta$-connected load with $Z_{\Delta} = -j\Omega$ through a transmission line with impedance of $j0.1\Omega$ per phase. The load is also connected to a $\Delta$-connected generator with $V_{a''b''} = 1\angle 0°$ through a second transmission line which also has an impedance of $j0.1\Omega$ per phase.

Find

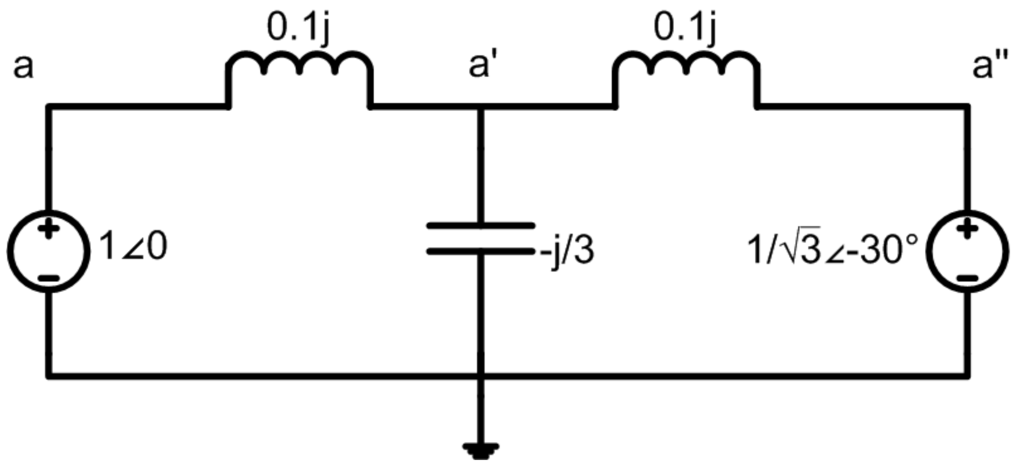
1. The load voltage $V_{a'b'}$
2. The total power supplied by each generator, $S_Y$ and $S_{\Delta}$

# Per Phase Example, cont’d




First convert the delta load and source to equivalent Y values and draw just the "a" phase circuit

# Per Phase Example, cont'd




To solve the circuit, write the KCL equation at a'

$$(V_a^{'} - 1\angle 0)(-10j) + V_a^{'}(3j) + (V_a^{'} - \frac{1}{\sqrt{3}}\angle -30°)(-10j) = 0$$

# Per Phase Example, cont'd

To solve the circuit, write the KCL equation at a'

$$(V_a^{'} - 1\angle 0)(-10j) + V_a^{'}(3j) + (V_a^{'} - \frac{1}{\sqrt{3}}\angle -30°)(-10j) = 0$$

$$(10j + \frac{10}{\sqrt{3}}\angle 60°) = V_a^{'}(10j - 3j + 10j)$$

$$V_a^{'} = 0.9\angle -10.9° \text{ volts} \qquad V_b^{'} = 0.9\angle -130.9° \text{ volts}$$

$$V_c^{'} = 0.9\angle 109.1° \text{ volts} \qquad V_{ab}^{'} = 1.56\angle 19.1° \text{ volts}$$

# Per Phase Example, cont'd

Conjugate

3 ×

$$S_{ygen} = 3V_a I_a^{*} = V_a \left( \frac{V_a - V_a^{'}}{j0.1} \right)^{*} = 5.1 + j3.5 \text{ VA}$$

$$S_{\Delta gen} = 3V_a^{''} \left( \frac{V_a^{''} - V_a^{'}}{j0.1} \right)^{*} = -5.1 - j4.7 \text{ VA}$$